

LOW ACRYLONITRILE NBR: PATENT LANDSCAPE AND PROCESS/COMPOSITION EVIDENCE

1. ABSTRACT

This report synthesizes two primary patents that intersect with low-acrylonitrile nitrile rubber chemistry, but they do so in different ways. US7592409B2 is primarily a controlled emulsion polymerization patent for styrene-based block copolymers that explicitly includes a nitrile-rubber block described as randomly distributed acrylonitrile and 1,3-butadiene; it is relevant as a process platform demonstrating aqueous microemulsion polymerization, in-situ emulsifier formation, and block-copolymer architectures containing NBR-like segments. US9624333B2 is more directly aligned with low-acrylonitrile NBR because it discloses ABS/graft-copolymer systems whose graft sheath contains acrylonitrile and/or methyl methacrylate, while the graft substrate is a butadiene-rich rubber prepared by aqueous emulsion polymerization. Across the two patents, the dominant synthesis motif is emulsion polymerization in water, often with potassium persulfate initiation, potassium hydroxide or bicarbonate buffering, and surfactant/soap stabilization. The first patent emphasizes controlled radical polymerization in microemulsion/latex form with in-situ surfactant generation and seed-latex block growth; the second emphasizes conventional aqueous emulsion polymerization of butadiene/styrene rubber substrates followed by agglomeration and grafting. For low-acrylonitrile NBR, the most important evidence is the explicit use of acrylonitrile as a comonomer in a rubber block or graft sheath and the ability to tune rubber composition through feed strategy and aqueous emulsion processing. However, neither patent is a dedicated low-ACN NBR grade patent; instead, they provide enabling process and composition windows that can support low-acrylonitrile rubber design, especially where acrylonitrile is present at controlled, relatively modest levels in a butadiene-rich copolymer. The landscape therefore points to aqueous emulsion routes, staged monomer addition, and composition control as the main levers for low-ACN NBR-related materials.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7592409B2
- Patent Title: Styrene acrylonitrile isoprene triblock copolymer

- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2009
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses controlled emulsion polymerization in an aqueous polymerization medium, including in-situ formation of an emulsifier from a latent surfactant and surfactant activator. It also describes block-copolymer synthesis using a vinyl-aromatic A block and a B block containing acrylonitrile/methacrylonitrile with a conjugated diolefin such as 1,3-butadiene or isoprene, enabling NBR-like block segments in latex form. - Latent surfactant amount: 0.2 grams — Example 1 - Styrene volume: 5 milliliters — Example 1 - Surfactant activator solution volume: 5 milliliters — Example 1 - Reaction vessel size: 2-gallon — Example 2 - Styrene charge: 1,000 grams — Example 2 - Oleic acid charge: 60.0 grams — Example 2 - Dibenzyltrithiocarbonate charge: 7.2 grams — Example 2 - RO water charge: 4,000 grams — Example 2 - Potassium persulfate charge: 40.0 grams — Example 2 - Tripotassium phosphate charge: 40.0 grams — Example 2 - Potassium hydroxide charge: 16.4 grams — Example 2 - Polymerization temperature: 65° C. — Example 2 - Conversion time: less than 1.5 hours — Example 2 - Solids after stripping: 20.6% — Example 2 - Mn: 54,000 — Example 2 - PDI: 1.17 — Example 2 - Reaction vessel size: 2-gallon — Example 3 - Styrene charge: 1,000 grams — Example 3 - Hexadecyl half ester of maleic acid charge: 60.0 grams — Example 3 - Dibenzyltrithiocarbonate charge: 7.2 grams — Example 3 - RO water charge: 4,000 grams — Example 3 - Potassium persulfate charge: 40.0 grams — Example 3 - Tripotassium phosphate charge: 40.0 grams — Example 3 - Potassium hydroxide charge: 16.4 grams — Example 3 - Polymerization temperature: 65° C. — Example 3 - Conversion time: less than 3 hours — Example 3 - Solids after stripping: 23.7% — Example 3 - Mn: 48,000 — Example 3 - PDI: 1.17 — Example 3 - RO water charge: 200 grams — Example 4 - RO water charge: 241 grams — Example 5 - RO water charge: 197.1 grams — Example 5 - RO water charge: 84.7 grams — Example 5 - RO water charge: 160.5 grams — Example 5 - RO water charge: 200 grams — Example 6 - Styrene charge: 1,000 grams — Example 6 - Oleic acid charge: 60.0 grams — Example 6 - PBN charge: 10.0 grams — Example 6 - Polymerization temperature: 75° C. — Example 6 - Conversion time: less than 2 hours — Example 6 - Solids after stripping: 22.7% — Example 6 - Mn: 25,000 — Example 6 - PDI: 1.80 — Example 6 - RO water charge: 241 grams — Example 7 - RO water charge: 197.1 grams — Example 7 - RO water charge: 84.7 grams — Example 7 - RO water charge: 241 grams — Example 7 - RO water charge: 197.1 grams — Example 7 - Styrene charge: 50 grams (0.48 moles) — Example 8 - Oleic acid charge: 3.0 grams (0.0106 moles) — Example 8 - PBN charge: 0.44 grams (0.002485 moles) — Example 8 - Distilled water charge: 200 grams — Example 8 - Potassium persulfate charge: 2.06 grams (0.00762 moles) — Example 8 - Tripotassium phosphate charge: 2.13 grams (0.01 moles) — Example 8 - Potassium hydroxide charge: 0.9 grams (0.014 moles) — Example 8 - Polymerization temperature: 65° C. — Example 8 - Conversion time: about 7 hours — Example 8 - Mn: 35,000 — Example 8 - Mw: 61,000 — Example 8 - PDI: 1.74 — Example 8 - Styrene charge: 200 grams

(1.923 moles) — Example 10 - Oleic acid charge: 12.0 grams (0.0424 moles) — Example 10 - PBN charge: 2.0 grams (0.0113 moles) — Example 10 - Distilled water charge: 500 grams — Example 10 - Potassium persulfate charge: 8.0 grams (0.296 moles) — Example 10 - Tripotassium phosphate charge: 8.0 grams (0.0377 moles) — Example 10 - Potassium hydroxide charge: 3.6 grams (0.056 moles) — Example 10 - Polymerization temperature: 75° C. — Example 10 - Conversion time: about 3.5 hours — Example 10 - Mn: 39,000 — Example 10 - Mw: 48,000 — Example 10 - PDI: 1.22 — Example 10 - PBN substitution: equimolar amount of PIN — Example 11 - Conversion time: about 7 hours — Example 11 - Mn: 41,000 — Example 11 - Mw: 70,000 — Example 11 - PDI: 1.70 — Example 11 - Seed latex charge: 100 grams of about 22% solids latex — Example 12 - Potassium persulfate charge: 0.5 grams (0.00184 moles) — Example 12 - Distilled water charge: 100 milliliters — Example 12 - n-Butyl acrylate charge: 22.0 grams (0.171 moles) — Example 12 - Polymerization temperature: about 65° C. — Example 12 - Weak exotherm: about 70° C. — Example 12 - Conversion time: about 2 hours — Example 12 - Mn: 45,000 — Example 12 - Mw: 77,000 — Example 12 - PDI: 1.70 — Example 12 - Oleic acid charge: 3.0 grams (0.0106 moles) — Example 13 - Benzaldehyde charge: 0.31 grams (0.00292 moles) — Example 13 - 15% aqueous solution of N-isopropylhydroxylamine: 1.4 grams (0.0028 moles) — Example 13 - Stirring time: 1 hour — Example 13 - Styrene charge: 50.0 grams (0.48 moles) — Example 13 - Distilled water charge: 200 grams — Example 13 - Potassium persulfate charge: 2.06 grams (0.00762 moles) — Example 13 - Tripotassium phosphate charge: 2.13 grams (0.01 moles) — Example 13 - Potassium hydroxide charge: 0.9 grams (0.014 moles) — Example 13 - Polymerization temperature: 75° C. — Example 13 - Conversion time: about 4 hours — Example 13 - Mn: 31,000 — Example 13 - Mw: 55,000 — Example 13 - PDI: 1.76 — Example 13 - Styrene charge: 50.0 grams (0.48 moles) — Example 14 - Dodecylbenzenesulfonic acid charge: 3.0 grams (0.0092 moles) — Example 14 - Distilled water charge: 1 milliliter — Example 14 - Sodium nitrite charge: 0.171 grams (0.002485 moles) — Example 14 - Distilled water charge: 200 grams — Example 14 - Potassium persulfate charge: 2.06 grams (0.00762 moles) — Example 14 - Tripotassium phosphate charge: 2.13 grams (0.01 moles) — Example 14 - Potassium hydroxide charge: 0.74 grams (0.0115 moles) — Example 14 - Polymerization temperature: 75° C. — Example 14 - Conversion time: about 3 hours — Example 14 - Mn: 97,000 — Example 14 - Mw: 147,000 — Example 14 - PDI: 1.51 — Example 14 - Styrene charge: 50 grams (0.48 moles) — Example 15 - Oleic acid charge: 3.0 grams (0.0106 moles) — Example 15 - 1,1-Diphenylethylene charge: 0.448 grams (0.002485 moles) — Example 15 - Distilled water charge: 200 grams — Example 15 - Potassium persulfate charge: 2.06 grams (0.00762 moles) — Example 15 - Tripotassium phosphate charge: 2.13 grams (0.01 moles) — Example 15 - Potassium hydroxide charge: 0.92 grams (0.0143 moles) — Example 15 - Polymerization temperature: 75° C. — Example 15 - Conversion time: about 2 hours — Example 15 - Mn: 51,000 — Example 15 - Mw: 92,000 — Example 15 - PDI: 1.80 — Example 15 - Styrene charge: 2,240 grams — Example 16 - Oleic acid charge: 134.4 grams — Example 16 - Dibenzyltrithiocarbonate charge: 16.0 grams — Example 16 - RO water charge: 3,808 grams — Example 16 - Potassium persulfate charge: 13.44 grams — Example 16 - Tripotassium phosphate charge: 89.6 grams — Example 16 - Potassium hydroxide charge: 36.96 grams — Example 16 - Polymerization temperature: 65° C. — Example 16 - Solids: 38.9% — Example 16 - Conversion time: less than 3 hours — Example 16 - Mn: 45,500 — Example 16 - PDI: 1.11 — Example 16 - Sodium nitrite charge: 0.69 grams (0.01 moles) — Example 17 - Oleic acid charge: 3.0 grams (0.0106 moles) — Example 17 - Styrene charge: 50

grams (0.48 moles) — Example 17 - Acetic acid solution: 1.0 grams (0.0167 moles) in 2 grams of water — Example 17 - Distilled water charge: 160 grams — Example 17 - Potassium persulfate charge: 2.0 grams (0.00739 moles) — Example 17 - Tripotassium phosphate charge: 2.0 grams (0.01 moles) — Example 17 - Potassium hydroxide charge: 2.35 grams (0.0367 moles) — Example 17 - Polymerization temperature: 75° C. — Example 17 - Conversion time: about 2.5 hours — Example 17 - Solids: 26.1% — Example 17 - Mn: 96,300 — Example 17 - Mw: 135,000 — Example 17 - PDI: 1.4 — Example 17 - Styrene charge: 50 grams (0.48 moles) — Example 18 - Oleic acid charge: 3.0 grams (0.0106 moles) — Example 18 - 1-chloro-1,1-diphenylethane: approximately 0.01 moles — Example 18 - Concentrated HCl: 10 milliliters — Example 18 - Toluene: 5 milliliters — Example 18 - Potassium persulfate charge: 2.0 grams (0.0074 moles) — Example 18 - Tripotassium phosphate charge: 2.0 grams (0.0094 moles) — Example 18 - Potassium hydroxide charge: 1.55 grams (0.0242 moles) — Example 18 - Distilled water charge: 200 grams — Example 18 - Polymerization temperature: 75° C. — Example 18 - Conversion time: about 4 hours — Example 18 - Solids: 22.7% — Example 18 - Mn: 40,000 — Example 18 - PDI: 1.37 — Example 18 - Styrene charge: 1,300 grams (12.5 moles) — Example 19 - Oleic acid charge: 78.0 grams (0.2756 moles) — Example 19 - Crude 1-chloro-1,1-diphenylethane charge: 39.0 grams — Example 19 - RO water charge: 2,470 grams — Example 19 - Potassium persulfate charge: 50.7 grams (0.1875 moles) — Example 19 - Tripotassium phosphate charge: 50.7 grams (0.2388 moles) — Example 19 - Potassium hydroxide charge: 29.5 grams (0.602 moles) — Example 19 - Polymerization temperature: 75° C. — Example 19 - Solids: 38.6% — Example 19 - Conversion time: about 5.5 hours — Example 19 - Mn: 32,000 — Example 19 - PDI: 1.85 — Example 19 - 1-hexene charge: 25 grams — Example 20 - n-Butyl acrylate charge: 25 grams — Example 20 - Dipyrindyl charge: 0.38 grams — Example 20 - Oleic acid charge: 4.0 grams — Example 20 - Ethyl 2-bromoisobutyrate charge: 0.47 grams — Example 20 - KOH solution: 1.7 grams of 85% KOH in 139 milliliters of distilled water — Example 20 - Copper sulfate pentahydrate solution: 60.9 grams of a 1% aqueous solution — Example 20 - Hydrazine hydrate: 2 drops — Example 20 - Reaction temperature: about 67° C. — Example 20 - Reaction time: 1.5 hours — Example 20 - Solids: 10.9% — Example 20 - ¹³C-NMR incorporation: 5.5 mole % 1-hexene — Example 20 - Medium & Water Role: The core reaction medium is an aqueous emulsion/microemulsion latex system. Water is explicitly used as the aqueous phase in emulsifier formation and polymerization, and the examples repeatedly describe water charges and latex formation. - low acrylonitrile content: Not disclosed in extracted evidence

Medium & Water Role - The core reaction medium is an aqueous emulsion/microemulsion latex system. Water is explicitly used as the aqueous phase in emulsifier formation and polymerization, and the examples repeatedly describe water charges and latex formation. - Core reaction medium: Aqueous emulsion / microemulsion latex polymerization

low acrylonitrile content - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 1: Demonstrated multiple latent surfactant/surfactant activator combinations that generate emulsions in styrene/water. - Example 2: Showed RAFT-controlled emulsion polymerization to a stable polystyrene latex with low PDI. - Example 3: Showed RAFT polymerization using an in-situ generated polymerizable surfactant and

stable latex formation. - Example 4: Demonstrated emulsion triblock copolymer formation from seed latex using water and butadiene/isoprene feeds. - Example 5: Demonstrated preparation of S-NBR-S triblock copolymer latexes from PS seed latexes.

Technical Relevance This patent is relevant because it explicitly discloses a nitrile-rubber block containing acrylonitrile and 1,3-butadiene and provides a water-based controlled emulsion route for making such architectures. Although not a dedicated low-ACN NBR grade patent, it shows how acrylonitrile-containing rubber segments can be synthesized in aqueous latex systems with controlled radical methods.

Patent 2

- Patent Number: US9624333B2
- Patent Title: Thermoplastic moulding materials based on acrylonitrile, styrene and butadiene
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses aqueous emulsion polymerization of a butadiene/styrene graft substrate followed by agglomeration and grafting to form ABS-type thermoplastic molding materials. The graft substrate is prepared in water with staged monomer feed, potassium persulfate initiation, potassium stearate emulsification, and sodium hydrogencarbonate buffering; the graft sheath then incorporates styrene and/or α -methylstyrene with acrylonitrile and/or methyl methacrylate. - Styrene content: 7 wt % — Example B1-1 - Reactor size: 150 L — Example B1-1 - Polymerization temperature: 67° C. — Example B1-1 - Monomer mixture charge: 43,120 g — Example B1-1 - Tert-dodecylmercaptan charge: 431.2 g — Example B1-1 - Potassium stearate charge: 311 g — Example B1-1 - Potassium persulfate charge: 82 g — Example B1-1 - Sodium hydrogencarbonate charge: 147 g — Example B1-1 - Water charge: 58,400 g — Example B1-1 - Conversion: 95% — Example B1-1 - Styrene content: 14 wt % — Example B1-2 - Styrene content: 21 wt % — Example B1-3 - Styrene content: 28 wt % — Example B1-4 - Total styrene content: 7 wt % — Table 1 - Total styrene content: 14 wt % — Table 1 - Total styrene content: 21 wt % — Table 1 - Total styrene content: 28 wt % — Table 1 - Core styrene content: 7 wt % — Table 1 - Gel content: 78.9 wt % — Table 1 - Gel content: 72.2 wt % — Table 1 - Gel content: 75.9 wt % — Table 1 - Gel content: 76.7 wt % — Table 1 - QI: 22 — Table 1 - QI: 29 — Table 1 - QI: 20 — Table 1 - QI: 26 — Table 1 - d_w: 88.4 nm — Table 1 - d_w: 104.0 nm — Table 1 - d_w: 87.9 nm — Table 1 - d_w: 82.2 nm — Table 1 - d_10: 72.3 nm — Table 1 - d_10: 90.5 nm — Table 1 - d_10: 72.6 nm — Table 1 - d_10: 69.9 nm — Table 1 - d_50: 90.0 nm — Table 1 - d_50: 105.8 nm — Table 1 - d_50: 88.1 nm — Table 1 - d_50: 84.3 nm — Table 1 - d_90: 101.1 nm — Table 1 - d_90: 114.8 nm — Table 1 - d_90: 98.8 nm — Table 1 - d_90: 93.4 nm — Table 1 - U: 0.32 — Table 1 - U: 0.23 — Table 1 - U: 0.30 — Table 1 - U: 0.28 — Table 1 - Mersolat H95 charge: 6.29 g — Agglomerating Copolymer

C-1 - Demineralized water charge: 1177.2 g — Agglomerating Copolymer C-1 - Potassium persulfate charge: 4.87 g — Agglomerating Copolymer C-1 - Demineralized water charge: 209.2 g — Agglomerating Copolymer C-1 - Ethyl acrylate charge: 211.2 g — Agglomerating Copolymer C-1 - Temperature increase: 60° C. to 80° C. — Agglomerating Copolymer C-1 - Ethyl acrylate feed: 1691.3 g — Agglomerating Copolymer C-1 - Potassium persulfate feed: 3.98 g — Agglomerating Copolymer C-1 - Demineralized water feed: 170.9 g — Agglomerating Copolymer C-1 - Mersolat H95 feed: 32.95 g — Agglomerating Copolymer C-1 - Methacrylamide feed: 90.6 g — Agglomerating Copolymer C-1 - Demineralized water feed: 1248.8 g — Agglomerating Copolymer C-1 - Feed time: 405 minutes — Agglomerating Copolymer C-1 - Mersolat H95 charge: 17.83 g — Comparative Example C-2 - Demineralized water charge: 1177.2 g — Comparative Example C-2 - Potassium persulfate charge: 4.87 g — Comparative Example C-2 - Demineralized water charge: 209.2 g — Comparative Example C-2 - Ethyl acrylate charge: 211.2 g — Comparative Example C-2 - Temperature increase: 60° C. to 80° C. — Comparative Example C-2 - Ethyl acrylate feed: 1691.3 g — Comparative Example C-2 - Potassium persulfate feed: 3.98 g — Comparative Example C-2 - Demineralized water feed: 170.9 g — Comparative Example C-2 - Mersolat H95 feed: 21.38 g — Comparative Example C-2 - Methacrylamide feed: 90.6 g — Comparative Example C-2 - Demineralized water feed: 1248.8 g — Comparative Example C-2 - Feed time: 405 minutes — Comparative Example C-2 - Solids content: 40.0 wt % — Comparative Example C-2 - Agglomeration yield: 65.8 wt % — Example 1 - Agglomeration yield: 65.7 wt % — Example 2 - Agglomeration yield: 64.3 wt % — Example 3 - Agglomeration yield: 67.8 wt % — Example 4 - Agglomeration yield: 61.8 wt % — Example 5 - Agglomeration yield: 65.0 wt % — Example 6 - Agglomeration yield: 67.6 wt % — Example 7 - Agglomeration yield: 70.9 wt % — Example 8 - d₅₀ value: 386 nm — Example 1 - d₅₀ value: 293 nm — Example 2 - d₅₀ value: 390 nm — Example 3 - d₅₀ value: 290 nm — Example 4 - d₅₀ value: 385 nm — Example 5 - d₅₀ value: 289 nm — Example 6 - d₅₀ value: 380 nm — Example 7 - d₅₀ value: 291 nm — Example 8 - SAN ratio: 75:25 styrene to acrylonitrile — Thermoplastic Copolymer a - Viscosity number: 64 ml/g — Thermoplastic Copolymer a - Concentration: 5 g/l in dimethylformamide — Thermoplastic Copolymer a - Measurement temperature: 20° C. — Thermoplastic Copolymer a - Melt flow rate MVR: 64 ml/10 min — Thermoplastic Copolymer a - Melt flow measurement temperature: 220° C. — Thermoplastic Copolymer a - Load: 10 kg — Thermoplastic Copolymer a - Extrusion temperature: 200° C. to 250° C. — Thermoplastic Molding Compounds - Extruder speed: 700 rpm — Thermoplastic Molding Compounds - Batch size: 4 kg — Thermoplastic Molding Compounds - Medium & Water Role: The core reaction medium is an aqueous emulsion latex system for the graft substrate and agglomerating copolymer. Water is also used later for dilution and latex handling, and the examples include demineralized water additions and aqueous latex processing. - low acrylonitrile content: 10 to 30 wt % of acrylonitrile and/or methyl methacrylate in the graft sheath (B2)

Medium & Water Role - The core reaction medium is an aqueous emulsion latex system for the graft substrate and agglomerating copolymer. Water is also used later for dilution and latex handling, and the examples include demineralized water additions and aqueous latex processing. - Core reaction medium: Aqueous emulsion polymerization / latex process

low acrylonitrile content - 10 to 30 wt % of acrylonitrile and/or methyl methacrylate in the graft sheath (B2)

Relevant Experimental Evidence - Example B1-1: Demonstrated aqueous emulsion polymerization of a butadiene/styrene graft substrate at 67° C. with 95% conversion. - Examples B1-2 to B1-4: Showed that increasing total styrene content from 14 to 28 wt % changes particle-size and gel metrics. - Agglomerating Copolymer C-1: Demonstrated emulsion polymerization of ethyl acrylate/methacrylamide copolymer in water for agglomeration. - Comparative Example C-2: Showed a similar agglomerating copolymer made with a different emulsifier loading and produced a 40.0 wt % latex solids product. - Example table data: Showed agglomerated graft substrates with d50 values around 289-390 nm and agglomeration yields around 61.8-70.9 wt %.

Technical Relevance This patent is directly relevant because it discloses a butadiene-rich rubber substrate made by aqueous emulsion polymerization and a graft sheath containing acrylonitrile, which is the key compositional motif for low-acrylonitrile NBR-related materials. The staged feed strategy, water-rich medium, and control of styrene/butadiene ratio provide practical levers for tuning rubber composition and morphology.

PATENT COMPARISON TABLE

Patent	Applicant	Priority	Medium & Water Role	low acrylonitrile content	Key Findings
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US7592409B2	Not disclosed	2009	The core reaction medium is an aqueous emulsion/microemulsion latex system. Water is explicitly used as the aqueous phase in emulsifier formation and polymerization, and the examples repeatedly describe water charges and latex formation.	Not disclosed in extracted evidence	Example 1: Demonstrated multiple latent surfactant/surfactant activator combinations that generate emulsions in styrene/
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US9624333B2	Not disclosed	2017	The core reaction medium is an aqueous emulsion latex system for the graft substrate and agglomerating copolymer. Water is also used later for dilution and latex handling, and the examples include demineralized water additions and aqueous latex processing.	10 to 30 wt % of acrylonitrile and/or methyl methacrylate in the graft sheath (B2)	Example B1-1: Demonstrated aqueous emulsion polymerization of a butadiene/styrene graft substrate at 67° C. with 95% con
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3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7592409B2 focuses on controlled aqueous emulsion/microemulsion polymerization with in-situ emulsifier formation and explicitly includes acrylonitrile-containing rubber blocks, whereas US9624333B2 uses conventional aqueous emulsion polymerization of a butadiene/styrene graft substrate followed by grafting with an acrylonitrile-containing sheath.
- US7592409B2 discloses block-copolymer architecture (A-B-A triblock) and controlled radical methods (RAFT/NMP-type conditions), while US9624333B2 discloses graft copolymer architecture with agglomerated rubber substrate and SAN-type sheath.

- Both patents rely on water as the polymerization medium, but US7592409B2 emphasizes microemulsion/latex formation from latent surfactant systems, while US9624333B2 emphasizes staged monomer feed, potassium stearate emulsification, and agglomeration of rubber particles.
- For acrylonitrile content, US7592409B2 only states that the B block may comprise acrylonitrile and methacrylonitrile with 1,3-butadiene or isoprene, without a numeric ACN range; US9624333B2 explicitly sets the graft sheath at 10 to 30 wt % acrylonitrile and/or methyl methacrylate.
- US9624333B2 provides the more direct low-ACN compositional guidance for NBR-related materials, while US7592409B2 is more of an enabling process patent for aqueous controlled polymerization of nitrile-rubber-containing blocks.

4. CONCLUSION

The evidence indicates that low-acrylonitrile NBR-related synthesis is most strongly supported by aqueous emulsion routes with staged monomer addition and surfactant/initiator control. US9624333B2 is the most compositionally relevant patent because it explicitly discloses an acrylonitrile-containing sheath at 10 to 30 wt % and a butadiene-rich emulsion-prepared rubber substrate, while US7592409B2 contributes a controlled microemulsion/block-copolymer methodology that can accommodate acrylonitrile/butadiene segments. Based on the extracted evidence, the most defensible synthesis direction for low-ACN NBR is water-based emulsion polymerization with controlled feed strategy, potassium persulfate initiation, and careful surfactant/soap stabilization; however, the exact low-ACN rubber grade range is not directly quantified in US7592409B2 and is only indirectly supported in US9624333B2 through the acrylonitrile-containing graft sheath.

5. REFERENCES

- US7592409B2 | Styrene acrylonitrile isoprene triblock copolymer | Not disclosed | US | 2009 | <https://patents.google.com/patent/US7592409B2/en>
- US9624333B2 | Thermoplastic moulding materials based on acrylonitrile, styrene and butadiene | Not disclosed | US | 2017 | <https://patents.google.com/patent/US9624333B2/en>